

Proposal for:
Cosimo Racco
Partner
RVI Group Ltd.
120 Brodie St N
Thunder Bay, ON
P7C 3S3

Proposal Date:
2023-03-29

Proposal Number:
QU-043

Reference
CA ON RVI MSSB

Understanding of project

RVI Group Ltd has requested for pricing on design assistance for a ground source heat pump (GSHP) system project located 5 km west of Blind River, ON. The building is a social services building and is approximately 12,500 ft². GEOoptimize will work with the engineer of record (RVI Group) to perform design assistance for the ground heat exchanger (GHX).

Scope of work

The scope of work included in this proposal includes:

- Review of Carrier HAP 8,760 hourly energy model provided by RVI Group Ltd. This will include review of building geometry, building occupancy, lighting, and ventilation schedules.
- Use of the energy model hourly loads to develop a ground coupled heat pump system option. Several iterations of the energy model will be developed to balance energy loads to the GHX as much as possible.
- Development of recommendations for building owners / architectural / engineering teams that will result in the most balanced energy loads possible for this facility. The recommendations will be reviewed with the owner / design teams and a final energy model will be used to develop a ground heat exchanger (GHX) model for the site.
- A model and technical explanation of the proposed GHX system.
- Assistance with heat pump and equipment selection.

Deliverables

Deliverables for this design review / study include:

- Recommendations for the owners / architect / engineers for changes to the building and / or mechanical systems that will help balance the energy loads to and from the GHX.
- Preliminary GHX design for the building site based on the land area available and discussion with GHX drilling contractors familiar with the area.

- Modeling and optimizing the hydraulic performance of the revised GHX design to:
 - Minimize pumping energy of the system
 - Maximize heat transfer between heat transfer fluid and the ground surrounding GHX piping
 - Facilitate efficient removal of air from the GHX borehole field and piping system
- Provide sketches of GHX and specifications
- Findings and recommendations will be summarized a presentation.

Pricing

Description	Rate	Qty	Line Total
Design Assistance	\$6,500.00	1	\$6,500.00
Subtotal			\$6,500.00
Proposal Total (CAD)			\$6,500.00

Terms

- Additional work out of scope will be charged at \$250/hr.
- Invoices will be submitted monthly for work completed to date.
- Payment is expected within 30 days of invoice.

Authorized signature – RVI Group Ltd.